

Robert Hart

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AI-native full-stack engineer designing systems that connect people and technology. 15 years of integration work as educator, researcher, and live experience designer — now building software where these domains converge.

Software Engineering Fellow — Fractal Tech, NYC

Feb 2026 – Present · Full-time fellowship (700+ hours; six-day weeks)

- Ship full-stack web applications (React, TypeScript, Express, MongoDB) in rapid sprint cycles
- Lead systems integration on a 4-person team: designed coordination process (kanban automation, dependency filtering, priority sorting & QA) that more than doubled team productivity on a week-long game build
- Architect real-time multiplayer systems using WebSocket broadcast patterns and server-authoritative state
- Build and operate multi-agent AI workflows: parallel dispatch, phase-sequenced builds, handoff protocols, memory management, and context engineering via custom agent skills

AI Engineering Intern — Stage 11 Agentics, NYC

Apr 2026 · Two-week sprint via Fractal Tech fellowship

- Contributed to an AI-related project at an early-stage startup; worked directly with the founder, shipping production code through daily-review, weekly-delivery cycles
- Owned full delivery cycle on assigned features: implementation, code-review response, deployment to production, QA, and onboarding documentation enabling teammates to run the system from a clean install

Freelance Vocal Coach & Creativity Coach · 2023 – 2026 · NYC

- Designed individualized developmental programs for 30+ clients, specializing in stage fright intervention and embodied self-awareness
- Built pedagogical frameworks grounded in developmental psychology (Kegan, Piaget) that directly inform the scaffolding logic of Weft, a developmental AI tutoring platform

Lead Researcher — DesignIntelligence · 2020 – 2021 · Atlanta/Remote

- Led research teams producing long-form strategic reports for C-suite leaders in architecture, engineering, and construction
- Synthesized complex systems-level analysis (healthcare demographics, pandemic economics, industry forecasting) for stakeholders operating at multiple levels of abstraction

Immersive Theater Designer & Director · 2024 – 2025 · NYC

- Created and directed live branching-narrative experiences for up to 75 simultaneous participants across multiple venues, coordinating 26 characters with interlocking storylines, live improv, and audience-driven plot outcomes
- Co-designed real-time state management systems tracking player and faction status throughout 5-hr productions

Projects

Weft — Developmental AI Tutoring Platform | TypeScript · Claude Code Skills · Multi-agent orchestration

- Designed and built a developmental AI tutoring system with 10+ agent skills encoding pedagogical logic: gap-type classification, learner state tracking, intervention matching, and spaced review
- Architected multi-agent orchestration with parallel dispatch, phase sequencing, and handoff protocols; system routes around its own blind spots to preserve human teacher authority

The Schelling Point — Real-Time Multiplayer Game | React · TypeScript · Express · WebSocket · Embedding

- Built a cooperative multiplayer game where 20+ players converge on shared words using embedding-space centroids as the scoring engine
- Redesigned game mechanics overnight (centroid-based convergence scoring) and shipped a working demo to 20 live players; first group won the game together

Languages & Frontend: TypeScript, JavaScript, React, HTML, CSS

Backend & Data: Node.js, Express, WebSocket, MongoDB, PostgreSQL, SQLite, REST API design

AI & Tooling: Claude Code, multi-agent orchestration, context engineering, Git, Vite, Vitest

Education: University of Georgia — BA, English Literature & Music Business